UCL
Risk Assessment

Summary

Reference: RA127618/1 — **Sign-off Status:** Authorised

Date Created:	16/03/2026
Confidential?	No
Assessment Title:	Offsite testing for MECH0073 Prototype
Assessment Outline:	We (Aerosol Extraction System group) plan to test our extractor prototype at Agfa Cambridge, which we are working in collaboration with. We plan to test our prototype on their experimental printer setup which is representative of the printers our prototype is designed for.

Area Responsible (for management of risks)

Division, School, Faculty, Institute:	Faculty of Engineering Sciences
Department:	Dept of Mechanical Engineering
Group/Unit:	All Groups/Units

Location of Risks Off-Site

Building:	
Area:	
Sub Area:	

Further Location Information:	515 Coldhams Lane, Cambridge, CB1 3JS Agfa
Is additional GM or HG approval required? Only relevant to specialist biological risk assessments (GMM2, GMM3, HG2, HG3, GM animals, GM plants) except GMM class 1.:	**Click SELECT to change <u>ONLY</u> if this is a GMM Class 2, GMM Class 3, HG2, HG3, GM animals or GM plants risk assessment**
Assessment Start Date:	24/03/2026
Review or End Date:	01/12/2026
Country Location:	
GMM Class 2, GMM Class 3 or Hazard Group 3 Risk Assessment :	
Relevant Attachments:	
Description of attachments:	
Location of non-electronic documents:	There are no other documents at this point.
Assessor(s):	Goh, Teik Beng
Approver(s):	Jose Castrejon Pita
Signed Off:	Jose Castrejon Pita (24/03/2026 14:01)

PEOPLE AT RISK (from the Activities covered by this Risk Assessment) *

Reference: RA127618/1 **Sign-off Status: Authorised**

1. General site hazards

Description of Activity:	Testing of prototype extractor at Agfa Cambridge on their test printer rig located in their workshop
List those managing this Activity and their competence:	

Hazard 1. Injury at site

	Existing Control Measures	Further Control Measures
Testing is conducted at the test printer, which is around the workshop section at Agfa	Instructions and safety briefs to be followed. Team to remain vigilant and aware of the environment during the visit. Suitable safety equipment and wear to be used during the visit, e.g. safety glasses, if required by the industry partner. Students and staff to follow any safety operating procedures. Students have also been to the site multiple times prior for technical visits and meetings. Staff to brief the students about expected standards and code of behaviour. Identify safety requirements at time of arrival. Identify fire exits.	

Hazard 2. Slips, Trip and Falls

Trips over objects around partner workshop.	**Existing Control Measures** Workshop area is generally kept organised and tidy, and well organised between pathways and working areas. Area is well lit. Students and staff are to remain spatially aware of their surroundings at all times.	**Further Control Measures**

Hazard 3. Manual handling of prototype and ancillary equipment

Strain from lifting prototype components, test fixtures, high speed camera with its associated setup and vacuum.	**Existing Control Measures** Most components and equipment are below 5kg. Parts heavier than 5kg are machined parts that are manufactured at Agfa. Transport of components and equipment are shared equally between students. Lifting aids used where available. Moving heavier objects at site assisted by industry partner when necessary. Brief correct lifting technique; avoid twisting while carrying; plan rest points if moving long distances.	**Further Control Measures**

Risk Level

With Existing Controls:

Risk Level	**A - Very Low / Trivial**	Likelihood Severity

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2. Prototype testing

Description of Activity:	Testing of prototype on test printer at Agfa Cambridge
List those managing this Activity and their competence:	

Hazard 1. Moving parts

Test printer has a high speed industrial conveyor belt at 1ms-1	**Existing Control Measures** Students are to stand sufficiently far away from test printer when in operation. Students are only to approach test area under the approval and supervision of industry partner. Emergency stop button is located close to test area.	**Further Control Measures**

Hazard 2. Suction

Extractor prototype applies suction over a small area over the moving belt, next to printer nozzles	**Existing Control Measures** Suction applied is low, and ducting is secured with guidance and approval of industry partner.	**Further Control Measures**

Hazard 3. Aerosols

Inhalation of mist, skin/eye contact with inks from test printer	**Existing Control Measures** Appropriate clothing worn, in addition to PPE provided by industry partner when necessary. Students are to follow instructions of staff and technicians of industry partner before, during and after testing.	**Further Control Measures**

Hazard 4. Noise from machinery		
Elevated noise from combined plant and test printer operation	**Existing Control Measures** Earbuds are available everywhere throughout industry partner site.	**Further Control Measures**

Hazard 5. Setup of prototype onto test printer		
Working space of extractor is limited, and is near machinery of the test printer	**Existing Control Measures** Test rig is turned off when adjusting extractor prototype. Maintain safe distance within working space from machinery when adjusting prototype. Setup and adjustment done under supervision of staff and technician(s) at industry partner. Clear and systematic experimental procedure is made and briefed between students and industry partner.	**Further Control Measures**

Hazard 6. Data acquisition		
Tripods, cameras, and stands creating obstructions within work space; damage to site equipment and/or persons if knocked Sensors attached to extractor prototype for gathering pressure data.	**Existing Control Measures** Camera setup placed away from main pathways. High speed camera and stand used is compact. Camera setup also placed away from vibrating parts to prevent dropping. Sensors are connected to a display away from test printer. Test printer is turned off when collecting and recording data.	**Further Control Measures**

Risk Level

With Existing Controls:

Risk Level	**B - Low / Tolerable**	Likelihood Severity